adding up all the matter—dark or otherwise—as they had naively presumed. There's no more matter running around the cosmos today than had ever been estimated by the observers.

Nobody had foreseen the dominating presence of cosmic dark energy, nor had anybody imagined it as the great reconciler of differences.

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So what is the stuff? Nobody knows. The closest anybody has come is to presume dark energy is a quantum effect—where the vacuum of space, instead of being empty, actually seethes with particles and their antimatter counterparts. They pop in and out of existence in pairs, and don't last long enough to be measured. Their transient existence is captured in their moniker: virtual particles. The remarkable legacy of quantum physics—the science of the small—demands that we give this idea serious attention. Each pair of virtual particles exerts a little bit of outward pressure as it ever so briefly elbows its way into space.

Unfortunately, when you estimate the amount of repulsive "vacuum pressure" that arises from the abbreviated lives of virtual particles, the result is more than $10^{120}$ times bigger than the experimentally determined value of the cosmological constant. This is a stupidly large factor, leading to the biggest mismatch between theory and observation in the history of science.

Yes, we're clueless. But it's not abject cluelessness. Dark energy is not adrift, with nary a theory to anchor it. Dark energy inhabits one of the safest harbors we can imagine: Einstein's equations of general relativity. It's the cosmological constant. It's lambda. Whatever dark energy turns out to be, we already know how to measure it and how to calculate its effects on the past, present, and future of the cosmos.

Without a doubt, Einstein's greatest blunder was having declared that lambda was his greatest blunder.

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And the hunt is on. Now that we know dark energy is real, teams of astrophysicists have begun ambitious programs to measure distances and the growth of structure in the universe using ground-based and space-borne telescopes. These observations will test the detailed influence of dark energy on the expansion history of the universe, and will surely